A Frisch-Waugh-Lovell Theorem for GMM

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Paper by Adam Dearing (2026)

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Motivation

- We often care about a few parameters but need to include many controls.
- Can we partial out those controls and still obtain the same GMM estimate?

Dearing's contribution

- Textbook FWL theorem is limited to OLS
- Dearing (2026) extends FWL to GMM (allows for a nonlinear component and any fixed positive definite weighting matrix).

Why this is useful

- Identification comes from the variation left after controlling for confounders.
- We can avoid repeatedly absorbing fixed effects during nonlinear estimation.

Basic setup

Consider the linear model

$$Y = X_1\lambda + X_2\beta + \xi,$$

where λ is the parameter of interest and X_2 are the controls.

Apply the textbook FWL theorem

1. Regress each column of X_1 on X_2 and save the residuals, $\widetilde{X}_1$.
2. Optional: Regress Y on X_2 and save the residuals, $\widetilde{Y}$.
3. Regress $\widetilde{Y}$ or Y on $\widetilde{X}_1$ to obtain the same $\widehat{\lambda}$.

Data: n i.i.d. observations

- $y_i \in \mathbb{R}$ is the outcome variable
- $x_i \equiv (x_{1i}, x_{2i})$, where $x_{1i} \in \mathbb{R}^{k_1}$ are variables of interest and $x_{2i} \in \mathbb{R}^{k_2}$ are control variables.
- $z_{1i} \in \mathbb{R}^{\ell_1}$ contains excluded instruments.

Parameters and instruments

$$\theta = (\lambda, \beta), \qquad Z = [\, Z_1 \;\; X_2 \,].$$

- λ is the parameter of interest. β contains the control coefficients.
- x_{1i} may be endogenous. The controls also serve as instruments.
- Assume X_2 is linearly independent (remove redundant controls).

Assumption 1

The model implies a shock/residuals of the form

$$\xi_i(\lambda, \beta) = \rho(y_i, x_i; \lambda) - x_{2i}'\beta.$$

- We can compute $\rho(y_i, x_i; \lambda)$ at any candidate λ.
- The coefficients we partial out enter through the additive term $x_{2i}'\beta$.
- $\rho(\cdot)$ can be nonlinear and can also depend on the controls.

The linear model is a special case

$$\rho(y_i, x_i; \lambda) = y_i - x_{1i}'\lambda.$$

The same setup also covers inverted demand and firm conduct models.

GMM moment conditions

Assume the instruments $z_i = (z_{1i}', x_{2i}')'$ satisfy

$$\mathbb{E}[z_i \, \xi_i(\lambda^*, \beta^*)] = 0.$$

Full GMM problem

$$(\widehat{\lambda}, \widehat{\beta}) = \arg\min_{\lambda,\beta} \underbrace{(\rho(\lambda) - X_2\beta)' Z W Z' (\rho(\lambda) - X_2\beta)}_{Q(\lambda,\beta)}.$$

Assumption 2

- Weighting matrix W is symmetric and positive definite.
- Q is twice continuously differentiable with a unique interior minimizer.

Partial out the controls

$$M_2 \equiv I - X_2(X_2'X_2)^{-1}X_2', \qquad \widetilde{Z}_1 \equiv M_2 Z_1, \qquad \widetilde{\rho}(\lambda) \equiv M_2 \rho(\lambda).$$

These are the residuals from annihilating X_2.

Rewrite the same GMM criterion

Let $A \equiv (X_2'X_2)^{-1}X_2'Z_1$. Then

$$Z = \underbrace{[\,\widetilde{Z}_1 \;\; X_2\,]}_{\equiv \widetilde{Z}} \underbrace{\begin{bmatrix} \mathrm{Id} & 0 \\ A & \mathrm{Id} \end{bmatrix}}_{\equiv \Gamma}, \qquad \overline{W} \equiv \Gamma W \Gamma' = \begin{bmatrix} \overline{W}_{11} & \overline{W}_{12} \\ \overline{W}_{21} & \overline{W}_{22} \end{bmatrix}.$$

Changing the instruments requires a corresponding change in the weights.

The weight on the remaining moments

$$W_R \equiv \overline{W}_{11} - \overline{W}_{12}\overline{W}_{22}^{-1}\overline{W}_{21}.$$

This is the Schur complement of $\overline{W}_{22}$, and it is positive definite.

Theorem 1 (FWL Theorem for GMM)

Under the stated assumptions, the full-model estimate also solves

$$\widehat{\lambda} = \arg\min_{\lambda} \widetilde{\rho}(\lambda)'\widetilde{Z}_1 W_R \widetilde{Z}_1'\widetilde{\rho}(\lambda).$$

- The estimate is identical in finite samples, including under misspecification.
- The reduced problem removes β and X_2.
- Note that because M_2 is idempotent and symmetric, $\widetilde{Z}_1'\widetilde{\rho}(\lambda) = \widetilde{Z}_1'\rho(\lambda)$

High level overview

- Full GMM problem is

$$(\widehat{\lambda}, \widehat{\beta}) = \arg\min_{\lambda,\beta}(\rho(\lambda) - X_2\beta)'ZWZ'(\rho(\lambda) - X_2\beta)$$

$$\implies \widehat{\lambda} = \arg\min_{\lambda}\min_{\beta}(\rho(\lambda) - X_2\beta)'ZWZ'(\rho(\lambda) - X_2\beta)$$

- Dearing proves that

$$\min_{\beta}(\rho(\lambda) - X_2\beta)'ZWZ'(\rho(\lambda) - X_2\beta) = \widetilde{\rho}(\lambda)'\widetilde{Z}_1 W_R \widetilde{Z}_1'\widetilde{\rho}(\lambda)$$

- See Appendix A.1

Random coefficients logit demand (Berry, Levinsohn, and Pakes, 1995)

Inverting market shares gives mean utility $\delta(\lambda^{RC})$. The shock is

$$\xi(\lambda, \beta) = \underbrace{\delta(\lambda^{RC}) - X_1 \lambda^{lin}}_{\rho(\lambda)} - X_2 \beta, \qquad \lambda = (\lambda^{RC}, \lambda^{lin}).$$

- λ^{RC} governs preference heterogeneity and substitution patterns.
- $X_2\beta$ can include many additive fixed effects.

Where the repeated work comes from

- At each candidate λ^{RC}, recompute δ and absorb the fixed effects.
- With multi-way fixed effects, this can take a long time.

Multi-way fixed effects

- Some models have tens or hundreds of thousands of multi-way fixed effects
- With non-linear term, we need to use numerical optimization to estimate λ
- Standard practice for each candidate λ involves:
  1. Calculate $\rho(\lambda)$
  2. Annihilate fixed effects (X_2) from $\rho(\lambda)$ and Z_1 to obtain $\widetilde{\rho}(\lambda)$ and $\widetilde{Z}_1$
- Annihilating fixed effects is done via method of alternating projections
- Method of alternating projections can take long time when multi-way fixed effects are correlated
- Dearing suggests to partial out X_2 from Z_1 at beginning and use that $\widetilde{Z}_1'\widetilde{\rho}(\lambda) = \widetilde{Z}_1'\rho(\lambda)$

A convenient form for the reduced weight

If $W = (Z'\Psi Z)^{-1}$, with Ψ symmetric positive semidefinite and the inverse defined,

$$\boxed{W_R = (\widetilde{Z}_1'\Psi\widetilde{Z}_1)^{-1}.}$$

Estimator	**Choice of Ψ**	**Reduced problem**
Linear OLS	Id, with $Z_1 = X_1$	OLS of $\widetilde{Y}$ on $\widetilde{X}_1$
Linear 2SLS	Id	2SLS using $\widetilde{Y}$, $\widetilde{X}_1$ and $\widetilde{Z}_1$
Step s of GMM	$\mathrm{diag}(\widehat{\xi}^2_{s-1})$	GMM with residualized instruments

For iterative GMM, the next weighting matrix depends on residuals from the previous estimate:

$$W_{R,s} = \left[\widetilde{Z}_1' \operatorname{diag}(\widehat{\xi}_{s-1}^2)\widetilde{Z}_1\right]^{-1},$$

where s is the step number.

Option 1: Use the full-model residuals $\widehat{\xi}_{s-1}$

These give exact equivalence at each step. Recover them using $\widehat{\beta}$, which also can be recovered.

Option 2: Use residuals from recentered GMM problem

- **Exact:** two-step GMM initialized with $W_1 = (Z'Z)^{-1}$.
- **Generally asymptotic:** other initial weights or further iterations, under correct specification and standard regularity conditions.

Interpretation

We can understand GMM through variation in the instruments left after controlling for X_2.

Computation

With many additive controls or fixed effects, residualizing instruments once can avoid repeated work inside a nonlinear optimizer.

Scope

- We need the additive linear control block and the corresponding reduced weight.
- Updating the weight requires care about which residuals we use.

Thank you

Discussion

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Prices, markups and marginal costs

Suppose the model implies

$$p_i = \Delta(y_i, x_i; \lambda) + c_i, \qquad c_i = x_{2i}'\beta + \xi_i.$$

The parameter λ governs conduct through the markup function Δ. Rearranging gives

$$p_i = \Delta(y_i, x_i; \lambda) + x_{2i}'\beta + \xi_i$$
$$\implies \xi_i = p_i - \Delta(y_i, x_i; \lambda) - x_{2i}'\beta$$

Use of the theorem

Residualize the excluded instruments on the additive cost controls once.

Use $\widetilde{Z}_1'\rho(\lambda)$ and W_R to estimate conduct without repeatedly absorbing those controls as the markup changes.

Main paper

Dearing, Adam (2026). *A Frisch-Waugh-Lovell Theorem for GMM*.
Working paper, January 3, 2026.

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Background discussed in the paper

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Hansen, Lars Peter (1982). "Large Sample Properties of Generalized Method of Moments Estimators." *Econometrica* 50, 1029–1054.